MODULE SOFTWARE DOCUMENTATION / CENG 396

STRATA Digital Twin & Analytics

Module 4 | Semih Çakır and Enes Taha Kılıç

Design baseline: 5 November 2026
Instructor: Erkan Zergeroğlu

Purpose. Turn time-stamped measurements into a live, measurement-driven structural view, synchronized plots and reproducible trial analysis.

Scope and delivery status

This is a proposed software design. The accompanying package contains documentation, a proposed internal sample schema, synthetic fixtures and a local stdout replay utility. A live dashboard, physical integration and measured performance results are not yet implemented.

What Module 4 owns

Data reception and validation; time mapping; one-axis base/floor visualization; displacement and acceleration charts; source-quality indicators; trial recording; peak/RMS metrics; qualified control OFF/ON comparison; exports.

What stays with the producer modules

M1 runs the motor control loop locally. M2 acquires and calibrates IMU signals. M3 calibrates the camera and tracks markers. M4 consumes their outputs and sends no motor command.

Independent development

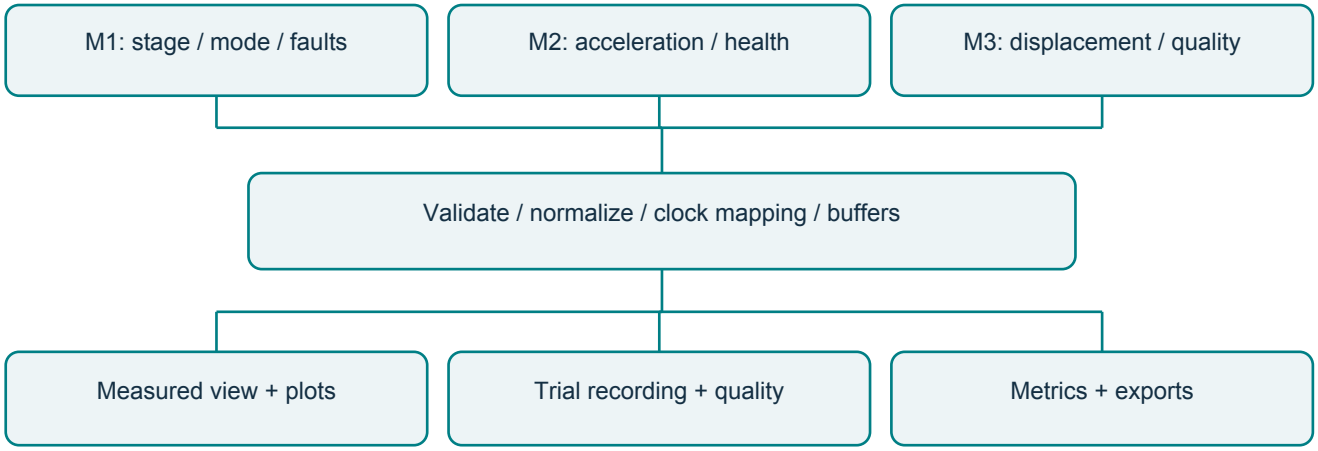
Mock or replayed streams allow progress without the physical setup. Synthetic records must remain visibly labelled and cannot be used as evidence of physical attenuation.

Editable source: Module4_Software_Design.md. Detailed companion files: Architecture.md, Data_Interfaces.md and Verification_Plan.md. This document is a module design, not the whole-group presentation.

01 / SYSTEM BOUNDARY

Inputs, outputs and data flow

Input	Meaning / use
M1: control status	Stage relative displacement, ON/OFF, limit/fault flags and timestamps. Display platform state and segment trials.
M2: IMU measurements	Calibrated acceleration, sensor identity, timestamps and health. Plot acceleration and compute per-sensor metrics.
M3: camera measurements	BASE and FLOOR_1-3 X displacement in mm, timestamps, confidence, validity and events. Drive measured structural motion.
Local setup / fixtures	Trial metadata, sensor map, time mapping; explicitly synthetic data or recorded replay.



Outputs: a measured structural view, time-series plots, stream/quality indicators, trial recordings, peak/RMS summaries, qualified OFF/ON comparisons and exports.

Transport and frontend framework are not finalized. Slow visualization must not block producer acquisition. The shared contract must be agreed with the other modules before live integration.

Receive and prepare measurements

Function IDs define responsibilities and interfaces; they are not claims of implemented functions.

Function	Input	Output / behavior
F01 load_config	Configuration	Validate sources, units, timing/quality policies and output paths; fail early on invalid settings.
F02 load_trial_metadata	Trial/source setup	Trial identity, source type, sensor map, mode, reference frames and clock maps; retain provenance.
F03 open_sources	Live/mock/replay settings	Independent source handles; one unavailable source does not stop the others.
F04 read_source_record	Source handle	Complete record or diagnostic; bounded buffers and maximum line size.
F05 validate_source_record	Record + adapter version	Typed accepted record or rejection; finite values, identities and units checked.
F06 normalize_record	Valid record + configuration	Internal channel samples with original timing, units, reference frame and validity; preserve raw input.
F07 map_time	Source time + approved map	Trial time with mapping provenance, or explicitly unsynchronized sample.
F08 update_stream_health	Arrival, sequence, status	Waiting/live/stale/disconnected plus measurement quality; connection does not imply valid data.
F09 buffer_samples	Mapped samples	Bounded per-channel data; deduplicate, record gaps and late arrivals without rewinding the view.

M3 local elapsed t_ms alone does not synchronize different devices. Keep source time, local receipt time and the approved trial-clock mapping distinct.

Display, analyze and preserve trials

Function	Input	Output / behavior
F10 build_view_state	Recent samples + health	Positions, quality and age; missing displacement is never replaced with zero.
F11 render_structure	View state + scale	Measured one-axis 2D view; visibly unavailable stale/invalid floors; labelled display exaggeration.
F12 update_plots	Buffers + time window	Shared trial-time plots with mm / m/s² units and gaps for invalid data; bounded refresh.
F13 start_trial	Setup + output location	Unique trial directory and provenance; refuse overwrite and mixed live/synthetic sources.
F14 record_sample	Raw / normalized records	Append measurements and events; recording failure visibly marks trial incomplete.
F15 compute_metrics	Channel + interval + policy	Peak, time-weighted RMS, coverage and exclusions; no unsupported metric.
F16 compare_trials	OFF / ON trials + metadata	Qualified side-by-side results; reject unknown mode, mixed provenance or insufficient coverage.
F17 export_trial	Records + metadata + metrics	CSV, JSON summary and human-readable report; preserve units, nulls and origin.
F18 close_trial	Resources + stop reason	Repeat-safe cleanup, completion status and diagnostics without hiding original failure.
F19 run_dashboard	Configuration + actions	Orchestrate input, rendering and recording independently; always release resources.

Only tools/replay.py is runnable now. It emits fixture records to stdout; it is not a network adapter, full validator, dashboard or implementation of this function table.

Keep meaning and provenance intact

M3 proposal already available

One measurement record per frame: schema_version, source/trial/frame identity, elapsed t_ms, timing_basis, module status, four markers and events. Markers carry name, ID, signed x_mm, confidence, valid and reason.

Invalid measurements use null and confidence 0 immediately. A marker_lost event follows the configured consecutive-invalid-frame threshold; do not wait for that event to show a gap. Reference loss invalidates all CV displacement.

Internal normalized sample

Fields	Purpose
source / trial_id / sequence / channel	Stable identity and deduplication key.
origin / recorded_origin	Distinguish live, synthetic and replay, including original recording provenance.
source_t_ms / timing_basis	Original source time and its declared meaning.
trial_t_ms / clock_map_id / synchronized	Approved time map or explicit null/unavailable alignment.
quantity / unit / reference_frame	Displacement in mm or acceleration in m/s²; never mix relative stage and fixed-support positions.
value / valid / confidence / reason	Finite numeric measurement or null, with validity and quality.

Proposed mapping: trial_t_ms = scale × source_t_ms + offset_ms. Record method, uncertainty and applicability. A reset/reconnect/map change creates an explicit segment. Arrival-only alignment is approximate and must be labelled.

M3 proposes optional LF-delimited JSON over TCP with trial_meta on connection; transport and metadata details need M4 agreement. M1/M2 wire schemas, rates and clocks remain unconfirmed. The internal schema does not impose a producer format.

Measured motion, defensible metrics

Live measured twin

Start with a front view of the fixed table, moving base and three floors. Use M3 positions relative to the fixed support and trial zero. IMU acceleration appears in its own plots; do not double-integrate it into position in the first implementation.

M1 stage position is relative to the moving base. Only compute stage_absolute = base_absolute + stage_relative when compatible reference frames and common timing are established. Never infer stage position from floor displacement.

Missing/stale measurements are visibly unavailable. A last-known outline may remain grey with its age. Optional short-gap display interpolation never enters metrics or bridges loss, clock, trial or mode boundaries.

Peak and time-weighted RMS

Peak: maximum absolute finite valid sample in the selected window.
RMS: square root of trapezoidal integral of squared samples divided by eligible duration. No mean removal.

For each valid adjacent pair: contribution = $0.5 \times (x_i^2 + x_{next}^2) \times dt$. Include only positive dt below the configured maximum gap. Coverage = eligible duration / requested duration. No bridging invalid samples; insufficient coverage makes RMS unavailable.

OFF/ON comparison

Reduction (%) = 100 × (OFF - ON) / OFF. Zero or near-noise OFF baselines make the ratio unavailable. Negative reduction indicates increased response. UNKNOWN mode is never silently treated as OFF.

Compare compatible channel/unit/sensor configurations, durations and disturbance context. Show base-input measurements alongside roof response. Manual shaking differs between runs; lower roof peak alone does not establish successful attenuation.

Synthetic data and real measurements must never be compared as physical evidence. Thresholds for coverage, gaps, noise and input comparability remain team decisions.

06 / INDEPENDENT VERIFICATION

Tests to run as features are built

These are acceptance cases, not a table of passed live-system tests.

Scenario	Expected evidence
Schema / source failure	Reject malformed/non-finite values; preserve null/validity. A missing IMU stream does not stop CV visualization.
Tracking / reference loss	Show FLOOR_3 gaps from first invalid frame; one loss/recovery transition. Reference loss makes all CV displacement unavailable.
Timing / sequence	Known clock map aligns signals; unmapped source stays separate. Deduplicate; late records do not rewind live view.
Known visual movement	-10 / 0 / +10 mm appears with correct X sign, scale and fixed floor height; synthetic label visible.
Constant signal	x=5 mm at t=0,1,2 s: peak=5 mm; RMS=5 mm; coverage=100%.
Irregular sampling	t=[0,1,3] s; x=[0,2,2] mm: integral=10 mm²s; RMS=sqrt(10/3) mm; peak=2 mm.
Missing interval	t=[0,1,2,3], x=[2,2,null,2]; max gap 1.1 s: eligible time=1 s; coverage=1/3 before coverage gate.
OFF / ON and origin	OFF=10, ON=6 gives 40%; ON=12 gives -20%; OFF=0 gives unavailable. Reject physical claims from synthetic pairs.
Export / load / failure	Reopen exported values with units/nulls/provenance; prevent overwrite; bound buffers; mark storage failures incomplete.

Detailed cases V01-V20 live in Verification_Plan.md. Record actual inputs, configuration, software revision, expected/actual output and pass/fail evidence during implementation.

07 / DELIVERY AND COORDINATION

Build independently, then integrate

Phase	Deliverable / status
Planning baseline - now	Design, input/output diagrams, interface proposals, internal schema, synthetic records and stdout replay utility.
Independent demo - 4 Dec target	Mock/replay ingestion, measured structure view, plots, quality indicators, recording and verified analytics.
Integration	Agreed producer schemas, sensor placement, clock mapping and transports; real stream validation and measured latency.
Final - 30 Dec target	Integrated OFF/ON trials, qualified reports, exports and user guide.

Decisions to agree with the group

Producer wire formats and metadata; transport/endpoints; shared trial identity; control-mode source; sensor placement/axes; units and reference frames; time origin/drift/uncertainty; measured rates and UI latency; freshness/gap/coverage limits; OFF/ON disturbance comparability.

Implementation choices still open

A 2D view is sufficient for the one-axis initial scope. 3D is optional. Frontend framework and deployment mode are not selected. Python is proposed for adapters/replay to align with M3, without prescribing the user interface stack.

Repository package and source baseline

Editable designs: docs/Digital Twin & Analytics/.
Schema, fixtures and utility: M4-digital-twin/.
All fixtures are synthetic. No live dashboard or physical test results are claimed.

Sources: repository README; M1 Physical Stabilization & Control PDF; M3 Computer Vision & Motion Tracking PDF; original Presentation_Group5.pdf. Reviewed baseline: a2a6a167e92f5e03a06b82f9e41210aad1ab35a9.

Module owners: Semih Çakır and Enes Taha Kılıç. Individual task division remains for the owners to agree. The whole-group presentation is outside this package.